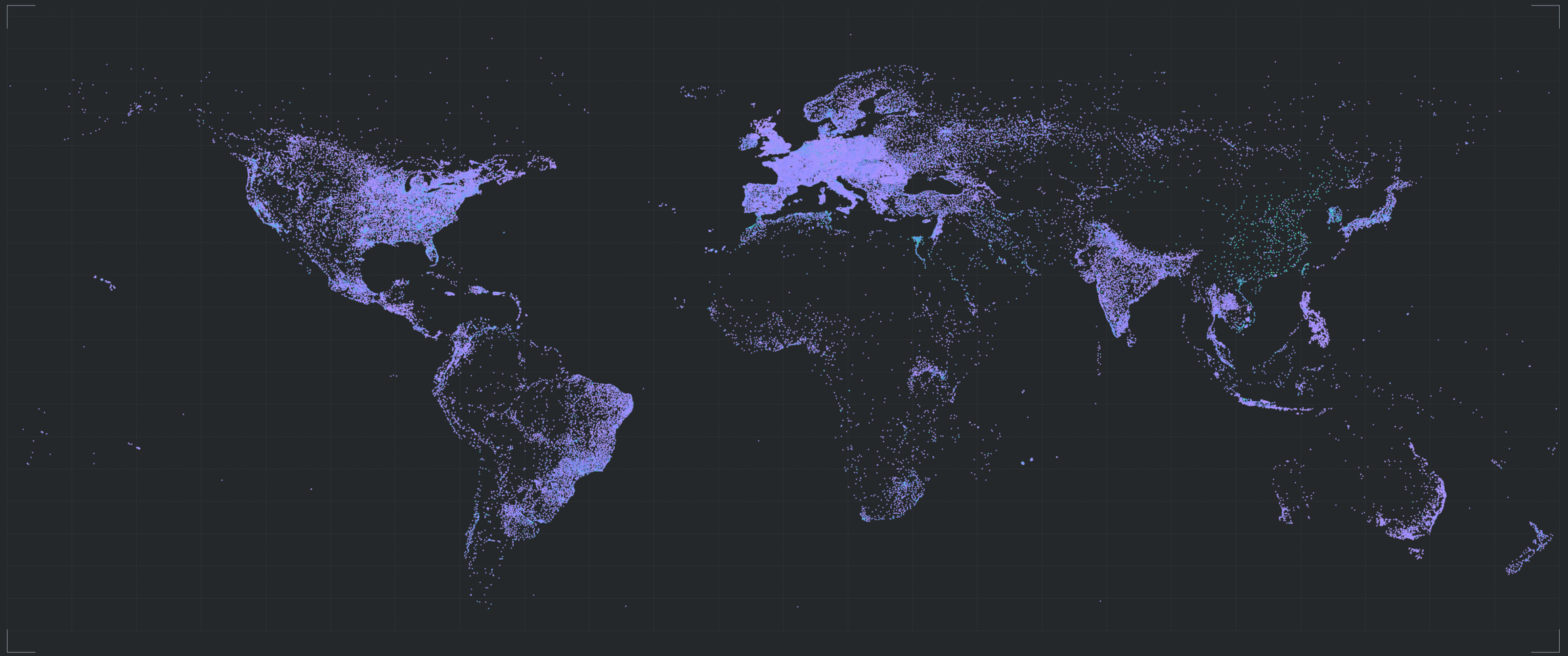
# Internet, emerging

IPv4 geolocation patterns / quantity at each database-assigned position

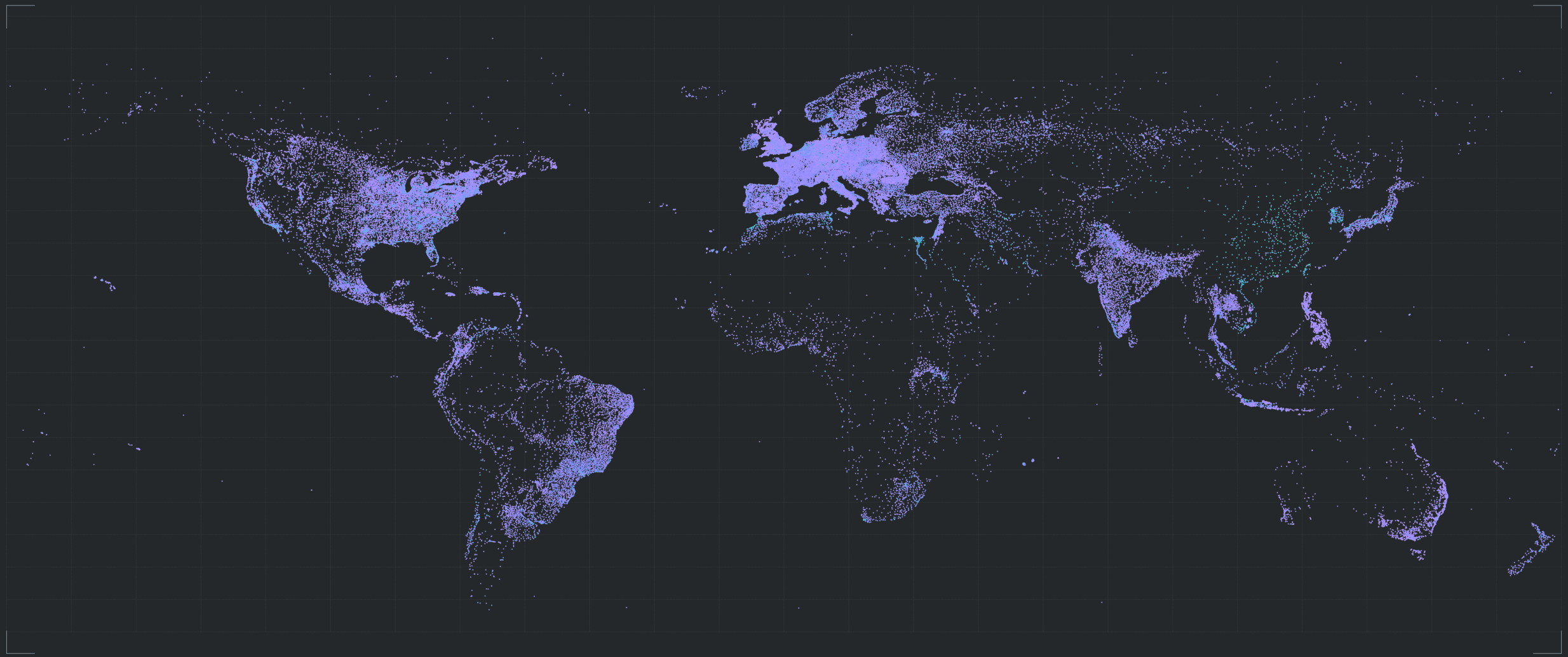


$10^3$ $10^4$ $10^5$ $10^6$ $10^7$ $10^8$

IPv4 addresses per location / logarithmic scale

METHOD / SOURCE NOTES

Source: IP2Location LITE DB11. Points are estimated geolocations, not individual servers or devices.



METHOD / SOURCE NOTES

IP2Location LITE DB11 / color encodes IPv4 addresses per location on the same logarithmic scale as the annotated map.

# Internet, as a pattern

One mark per distinct geolocation / monochrome study

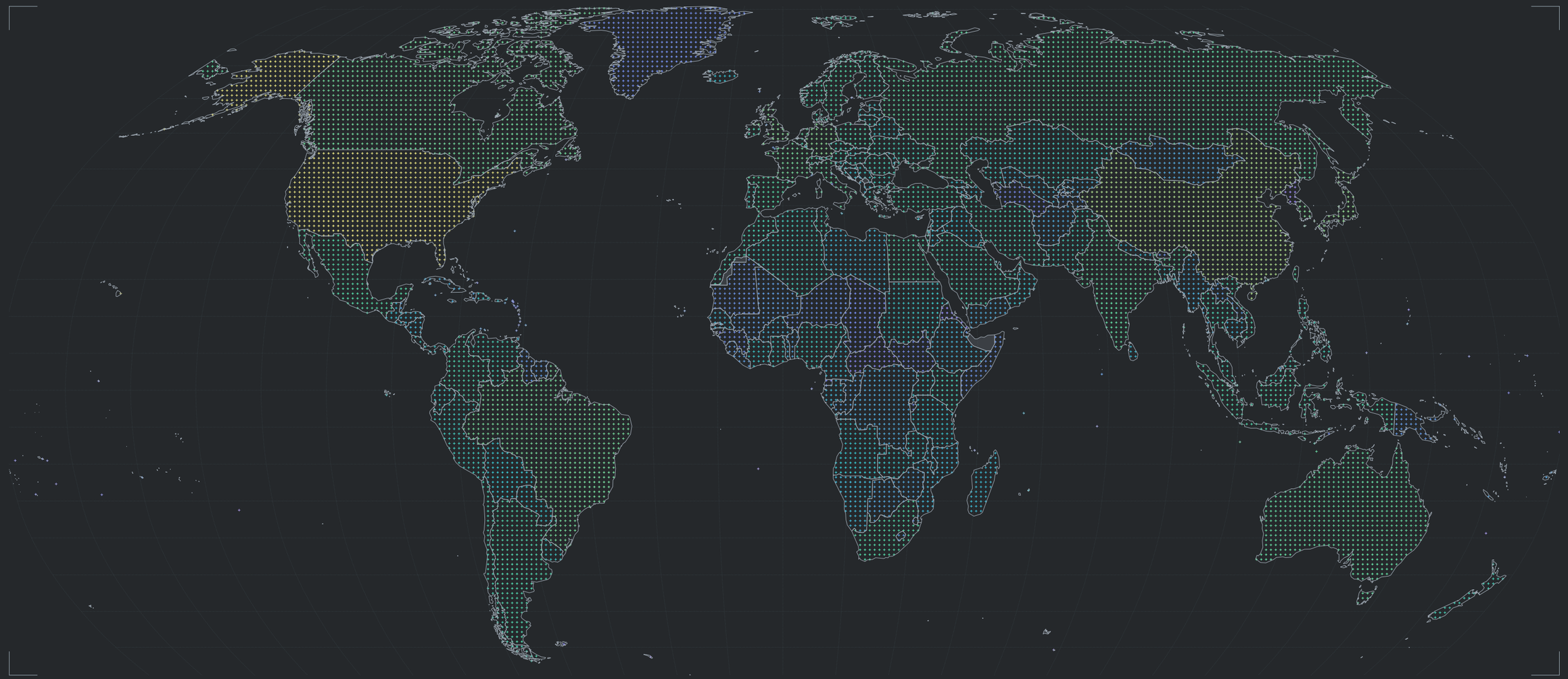


METHOD / SOURCE NOTES

Source: IP2Location LITE DB11. Marks have equal size; this version does not encode address quantities.

# Unequal address space

Geolocated IPv4 totals by country

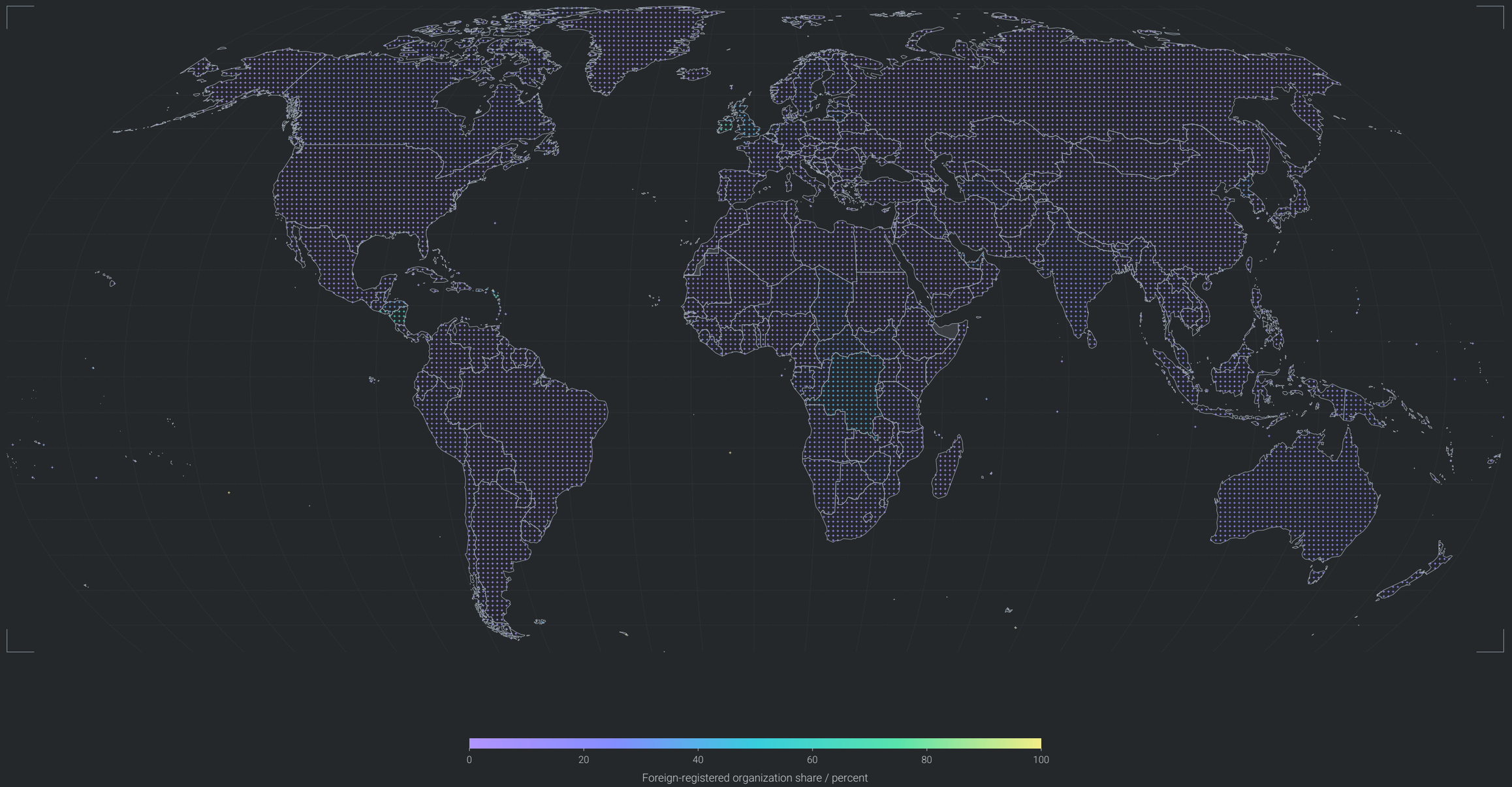



METHOD / SOURCE NOTES

Sources: IP2Location LITE DB11; Natural Earth. Hatch color encodes national totals; hatch density is constant.
Medium grey: no matched national data. Locations are estimates, not verified server positions.

# Across administrative borders

Foreign-registered origin-AS organizations / national share of observed IPv4

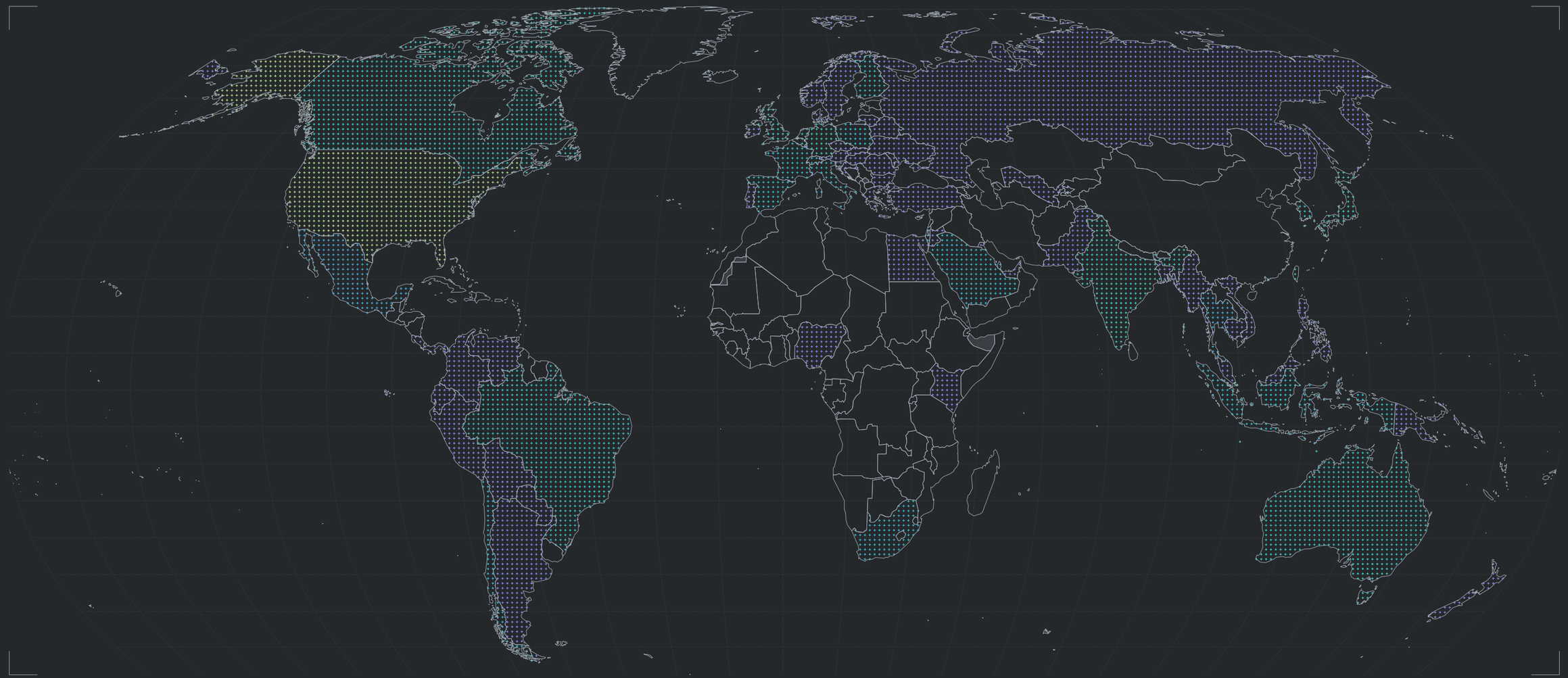



METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Constant-density hatching.
Unmarked dark: zero share; medium grey: no denominator. Unknown and ambiguous assignments remain in the denominator. Registration is not ownership.

# Google

Published IPv4 footprint / estimated country totals



$10^3$ $10^4$ $10^5$ $10^6$ $10^7$

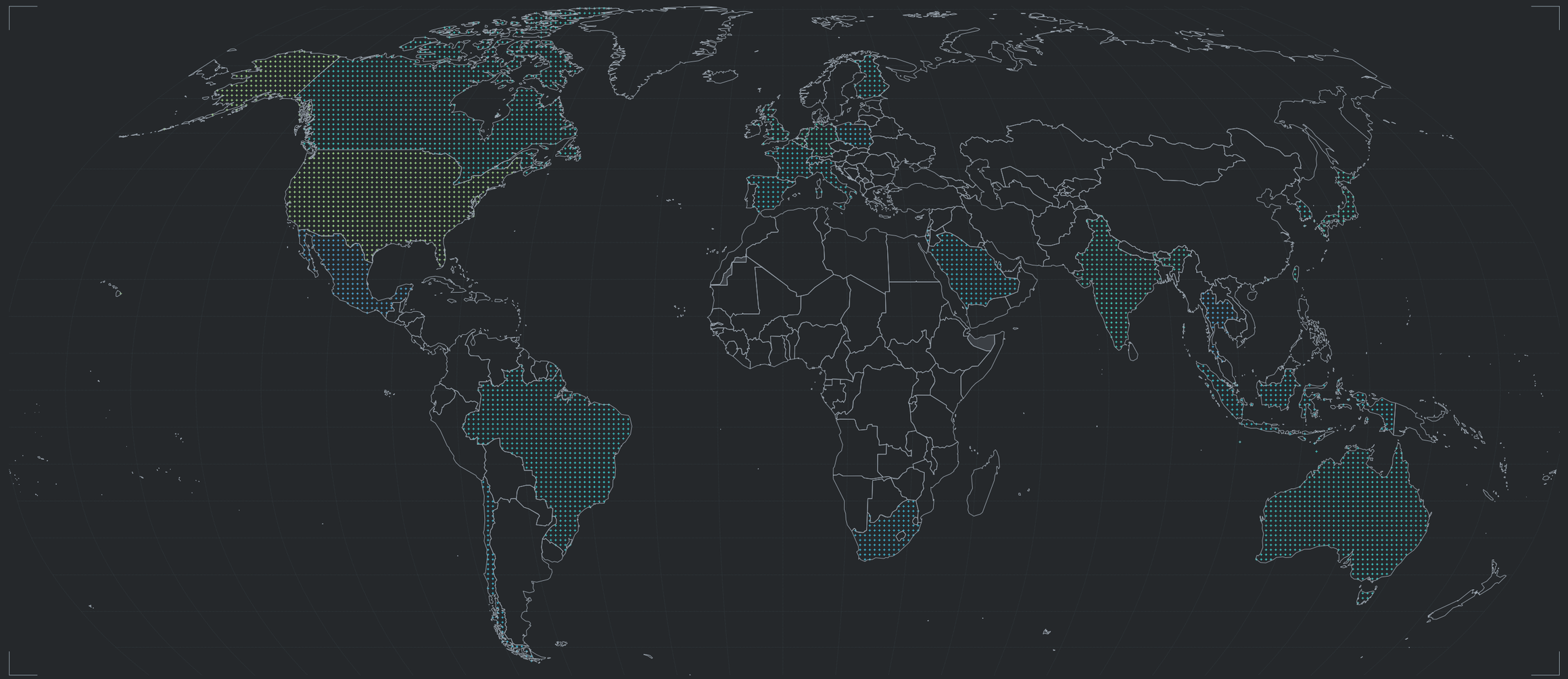
IPv4 addresses per country / shared logarithmic scale

METHOD / SOURCE NOTES

Sources: official provider lists; IP2Location LITE DB11; Natural Earth. Hatch color shows national counts; density is constant.
Unmarked dark: zero in this list; medium grey: no geolocation denominator. List scopes differ; Google Cloud overlaps Google.

# Google Cloud

Published IPv4 footprint / estimated country totals



$10^3$ $10^4$ $10^5$ $10^6$ $10^7$

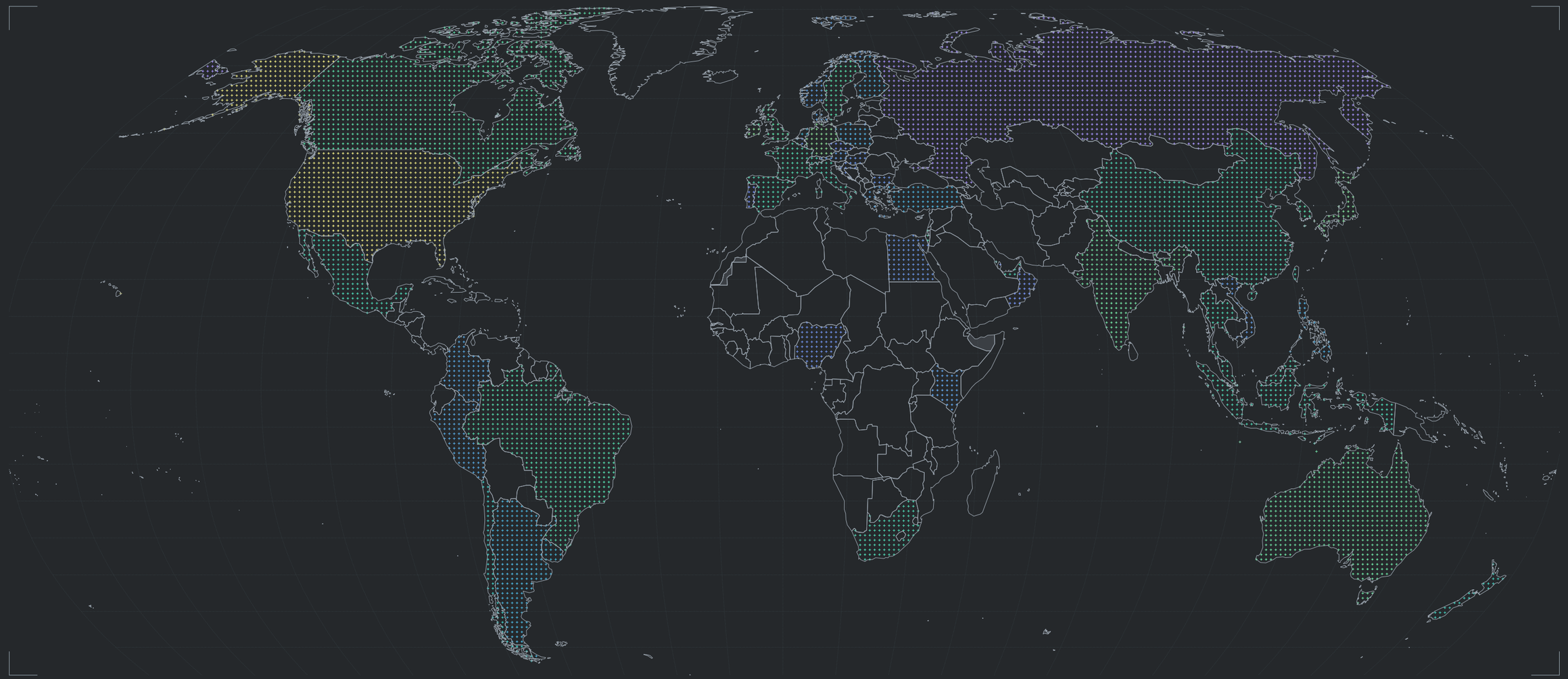
IPv4 addresses per country / shared logarithmic scale

METHOD / SOURCE NOTES

Sources: official provider lists; IP2Location LITE DB11; Natural Earth. Hatch color shows national counts; density is constant.
Unmarked dark: zero in this list; medium grey: no geolocation denominator. List scopes differ; Google Cloud overlaps Google.

# AWS

Published IPv4 footprint / estimated country totals



$10^3$ $10^4$ $10^5$ $10^6$ $10^7$

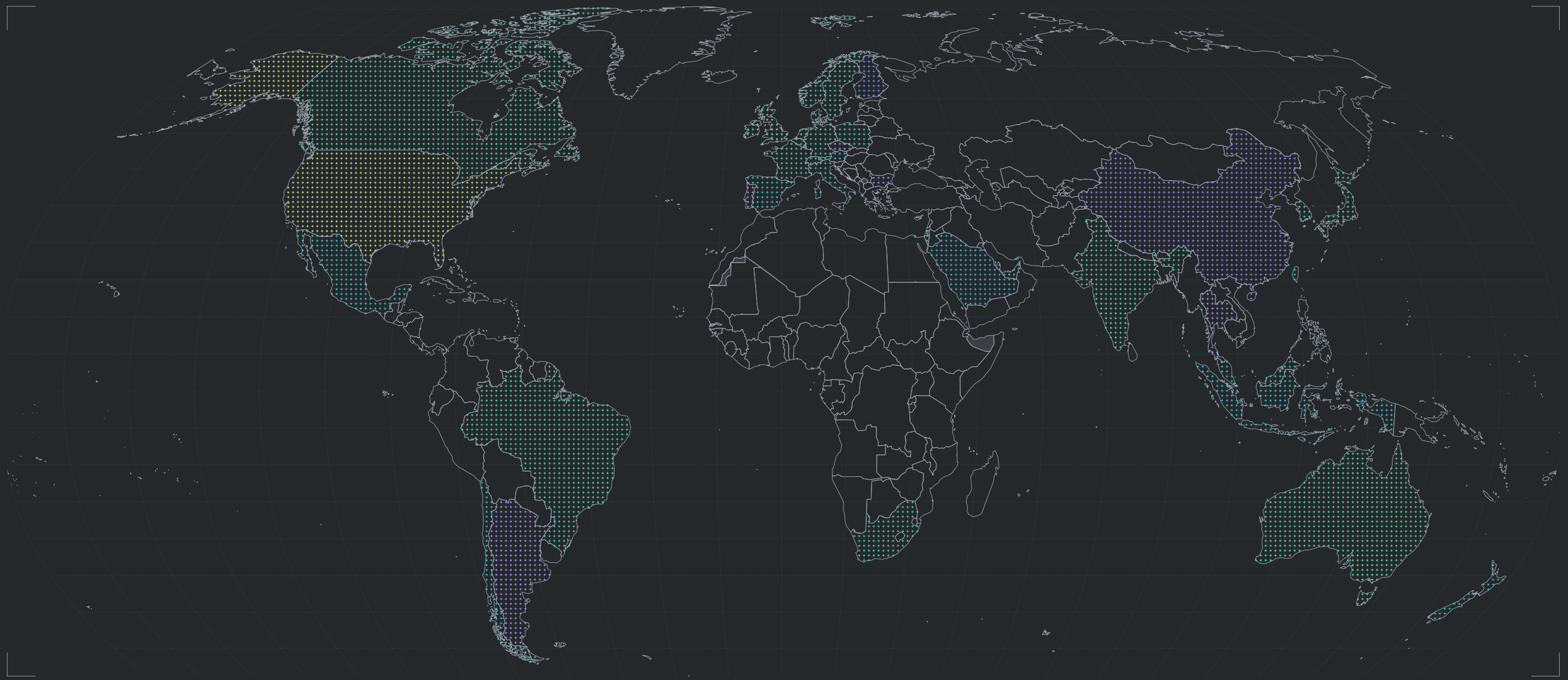
IPv4 addresses per country / shared logarithmic scale

METHOD / SOURCE NOTES

Sources: official provider lists; IP2Location LITE DB11; Natural Earth. Hatch color shows national counts; density is constant.
Unmarked dark: zero in this list; medium grey: no geolocation denominator. List scopes differ; Google Cloud overlaps Google.

# Azure Public

Published IPv4 footprint / estimated country totals



$10^3$ $10^4$ $10^5$ $10^6$ $10^7$

IPv4 addresses per country / shared logarithmic scale

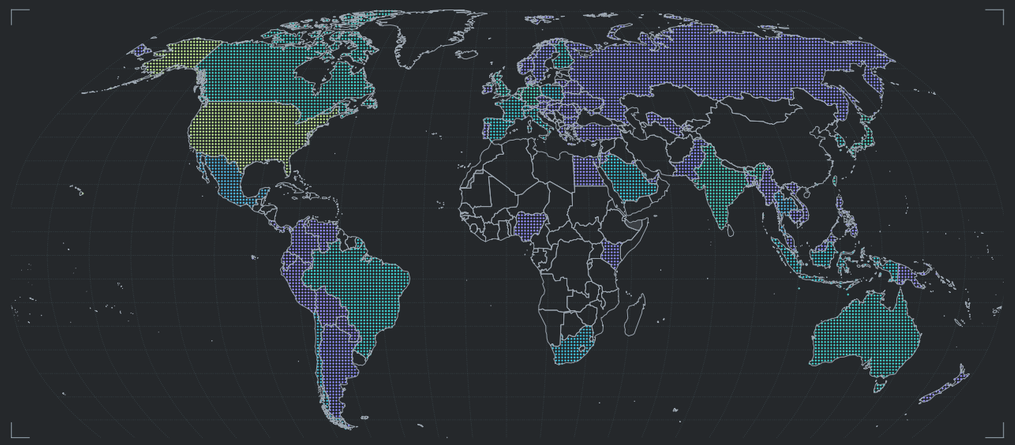
METHOD / SOURCE NOTES

Sources: official provider lists; IP2Location LITE DB11; Natural Earth. Hatch color shows national counts; density is constant.
Unmarked dark: zero in this list; medium grey: no geolocation denominator. List scopes differ; Google Cloud overlaps Google.

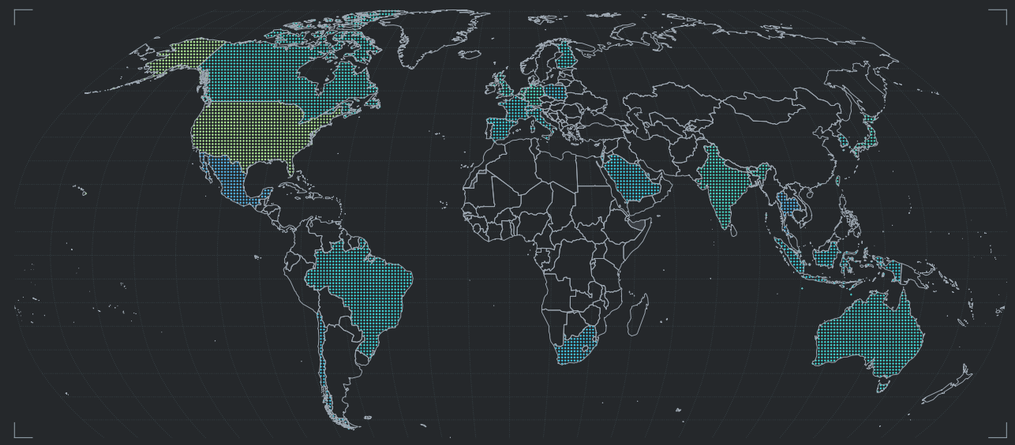
# Published corporate footprints

National totals / common color scale / different source scopes

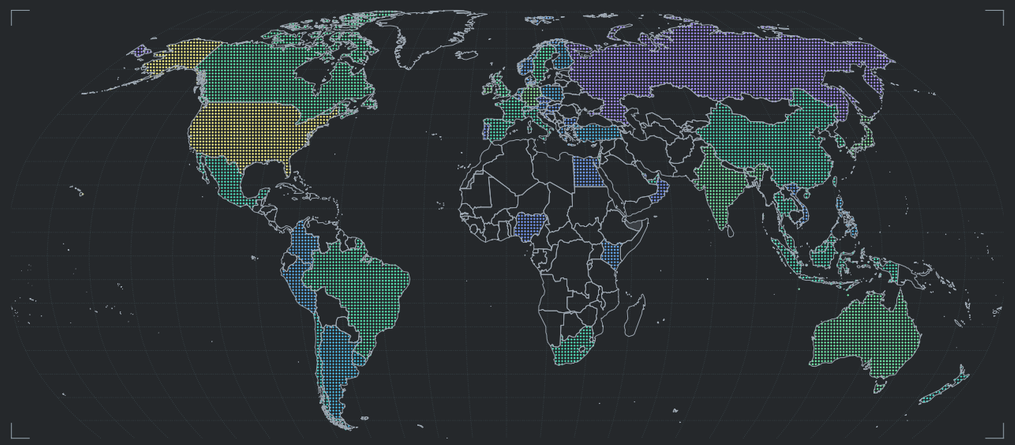
A / Google



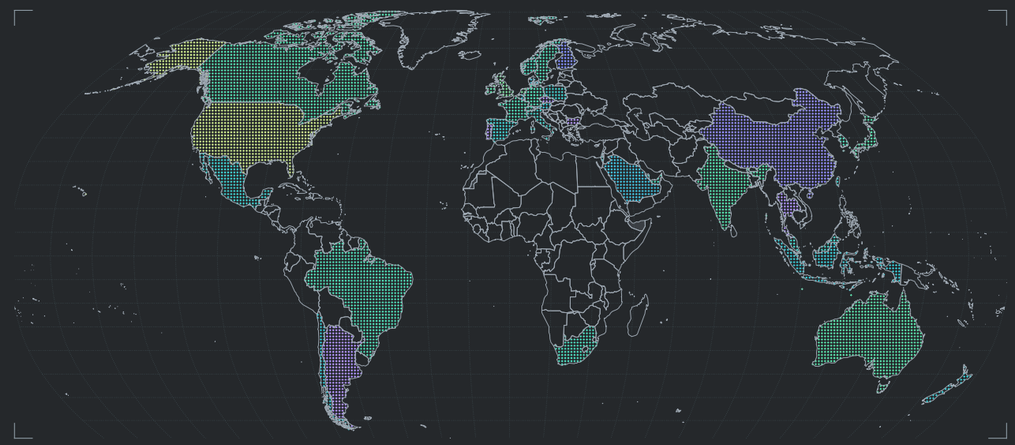
B / Google Cloud

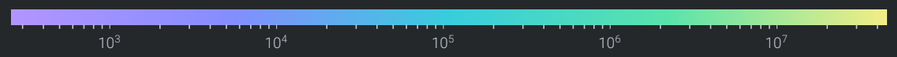


C / AWS



D / Azure Public





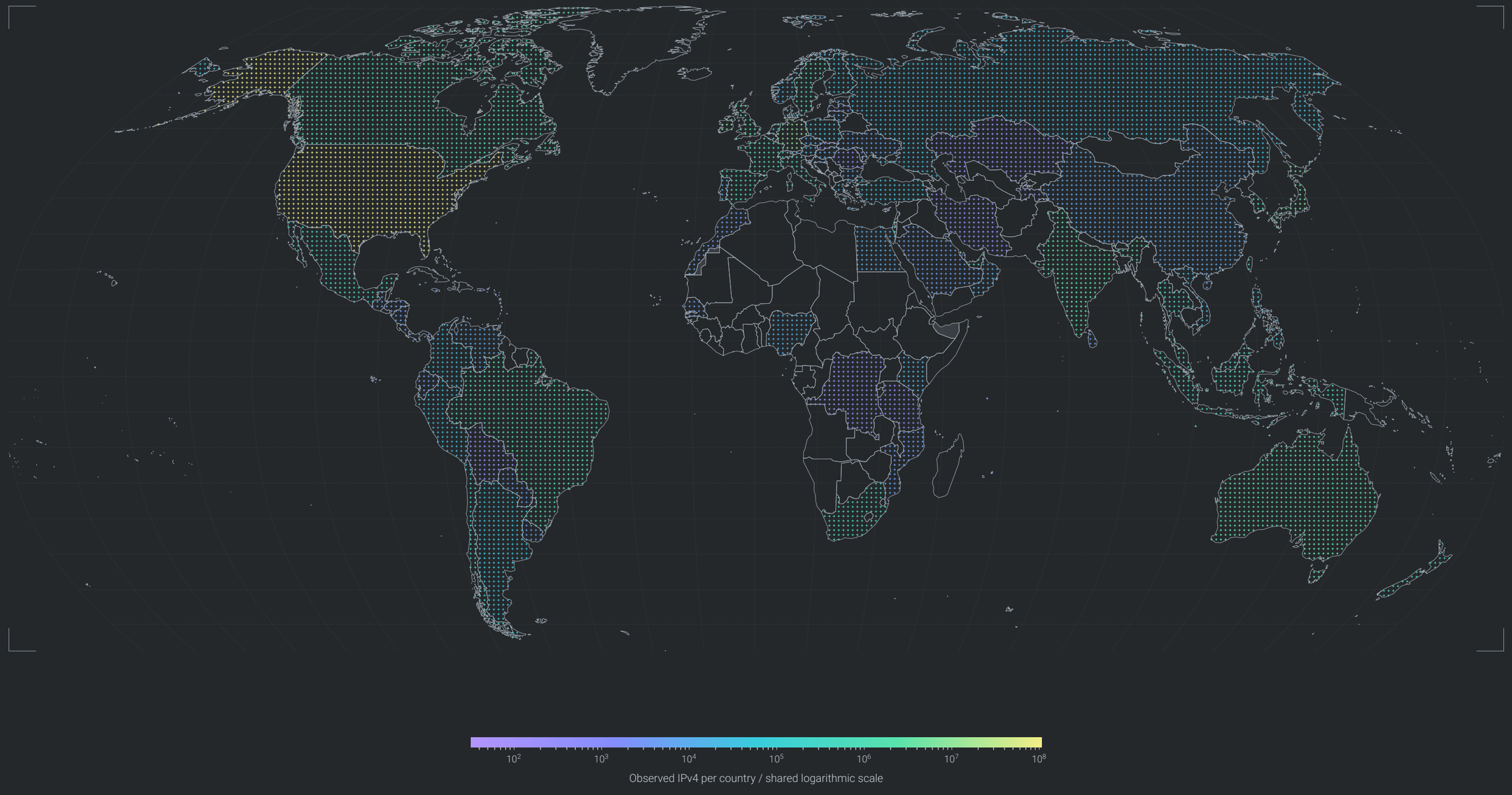

IPv4 addresses per country / logarithmic scale

METHOD / SOURCE NOTES

Sources: official provider lists; IP2Location LITE DB11; Natural Earth. Hatch color shows national counts; density is constant.
Unmarked dark: zero in this list; medium grey: no geolocation denominator. List scopes differ; Google Cloud overlaps Google.

# Amazon

Selected corporate networks / BGP-observed IPv4 by country

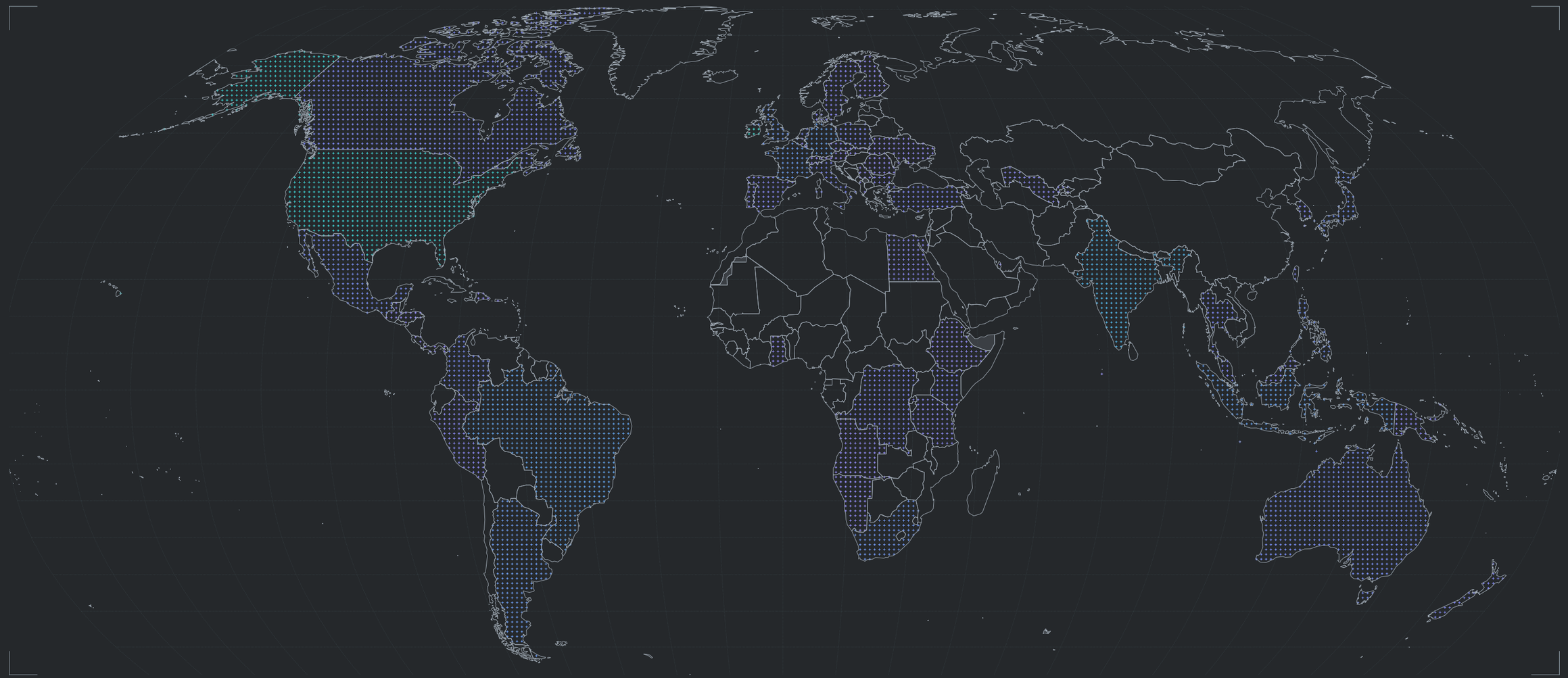



METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Same BGP method for all companies.
Primary organizations only. Hatch color: national IPv4 count. Unmarked dark: zero within selection; medium grey: no denominator. Not ownership.

# Meta

Selected corporate networks / BGP-observed IPv4 by country



$10^2$ $10^3$ $10^4$ $10^5$ $10^6$ $10^7$ $10^8$

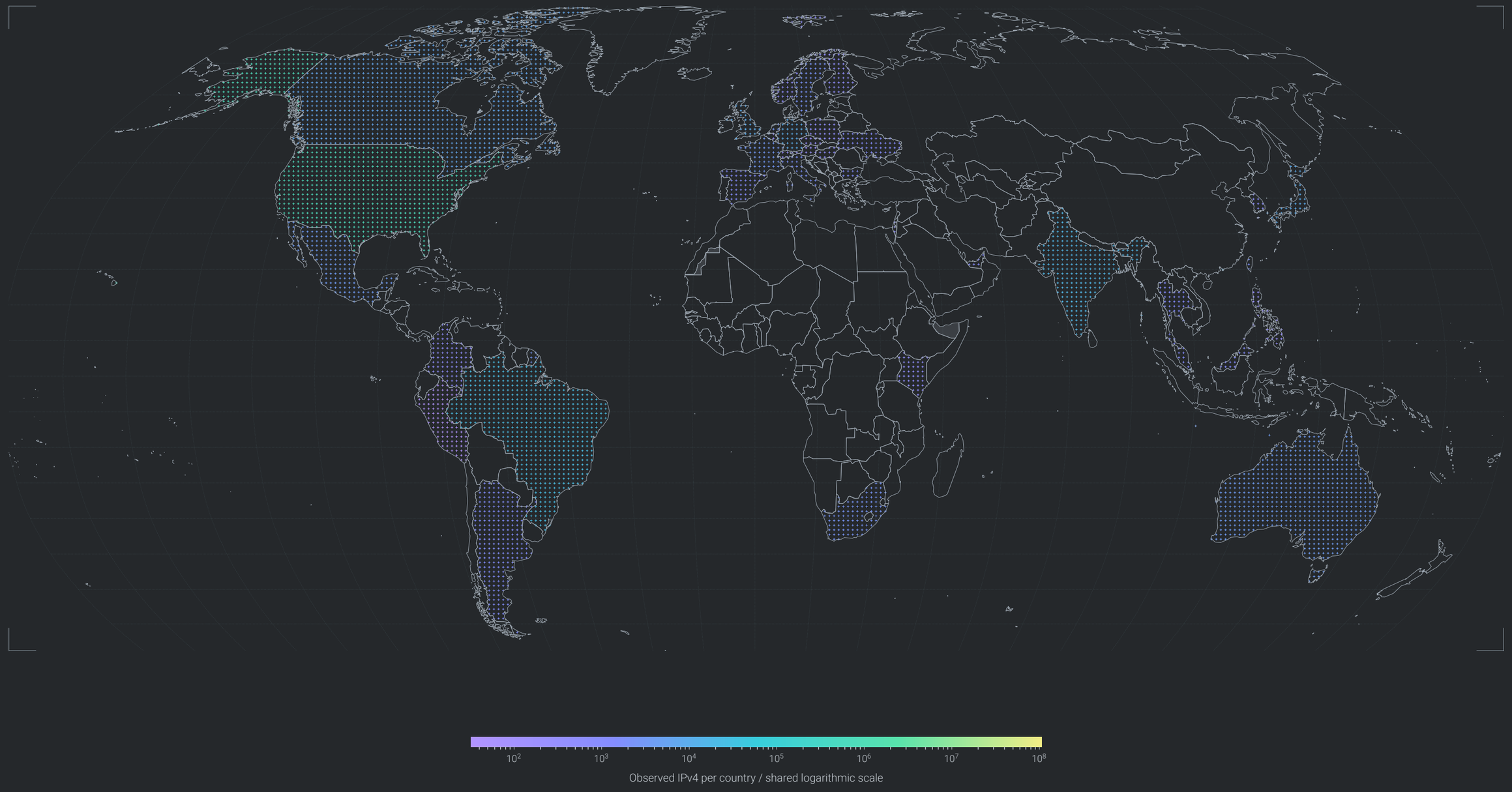
Observed IPv4 per country / shared logarithmic scale

METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Same BGP method for all companies.
Primary organizations only. Hatch color: national IPv4 count. Unmarked dark: zero within selection; medium grey: no denominator. Not ownership.

# Google

Selected corporate networks / BGP-observed IPv4 by country



$10^2$ $10^3$ $10^4$ $10^5$ $10^6$ $10^7$ $10^8$

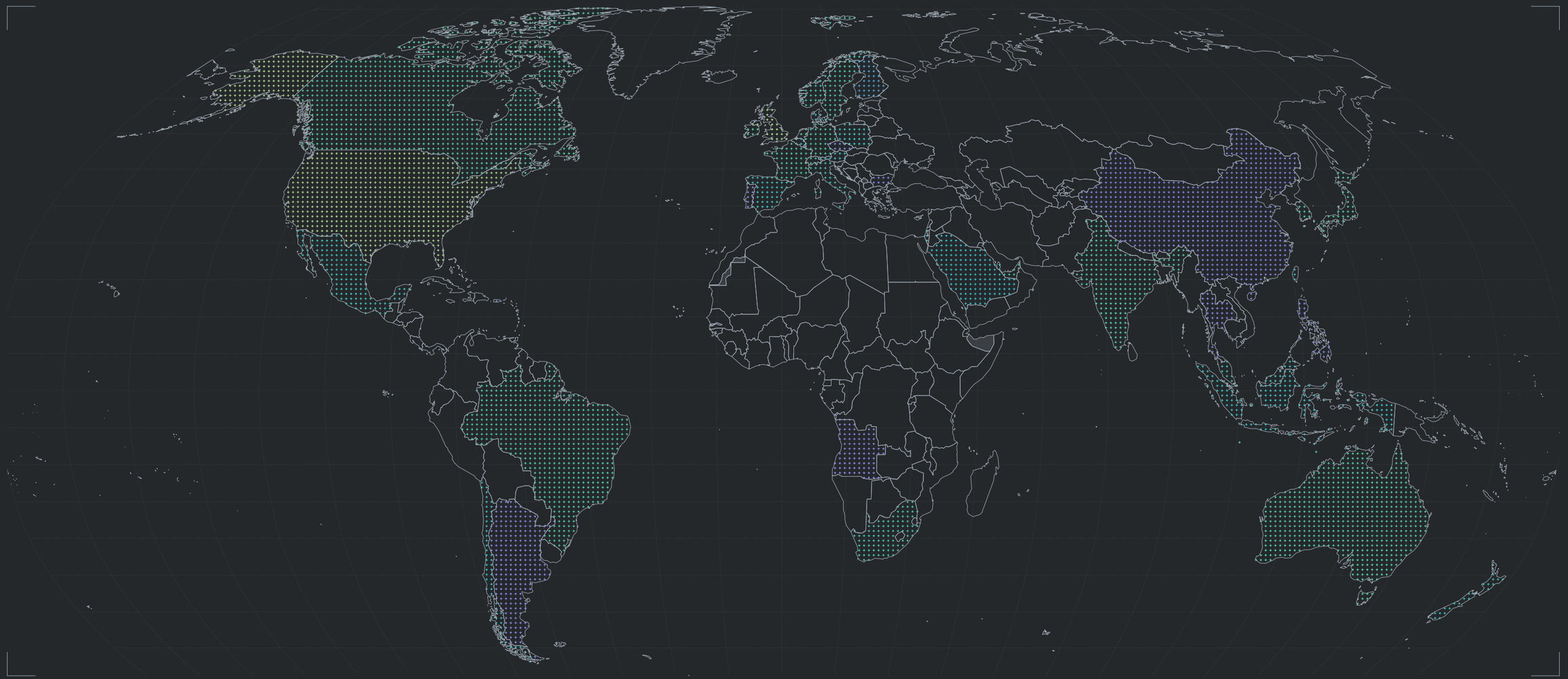
Observed IPv4 per country / shared logarithmic scale

METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Same BGP method for all companies.
Primary organizations only. Hatch color: national IPv4 count. Unmarked dark: zero within selection; medium grey: no denominator. Not ownership.

# Microsoft

Selected corporate networks / BGP-observed IPv4 by country



$10^2$ $10^3$ $10^4$ $10^5$ $10^6$ $10^7$ $10^8$

Observed IPv4 per country / shared logarithmic scale

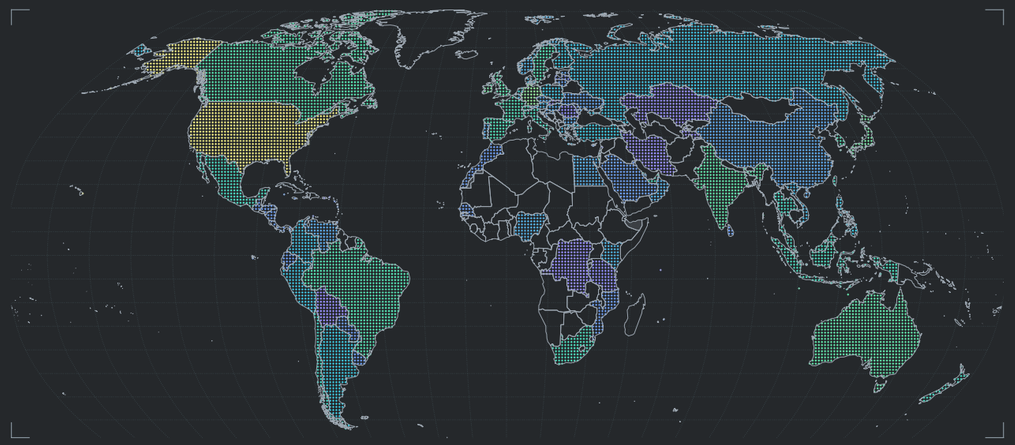
METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Same BGP method for all companies.
Primary organizations only. Hatch color: national IPv4 count. Unmarked dark: zero within selection; medium grey: no denominator. Not ownership.

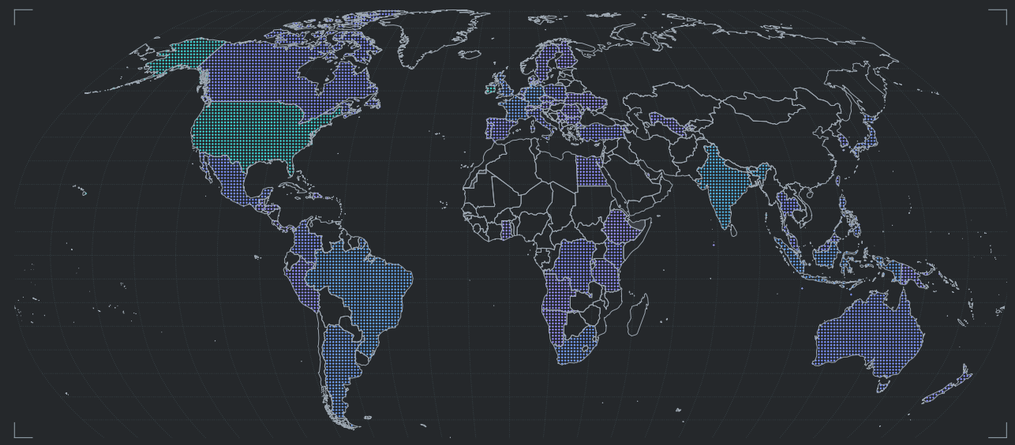
# Four companies, one routing lens

Amazon / Meta / Google / Microsoft - selected primary organizations

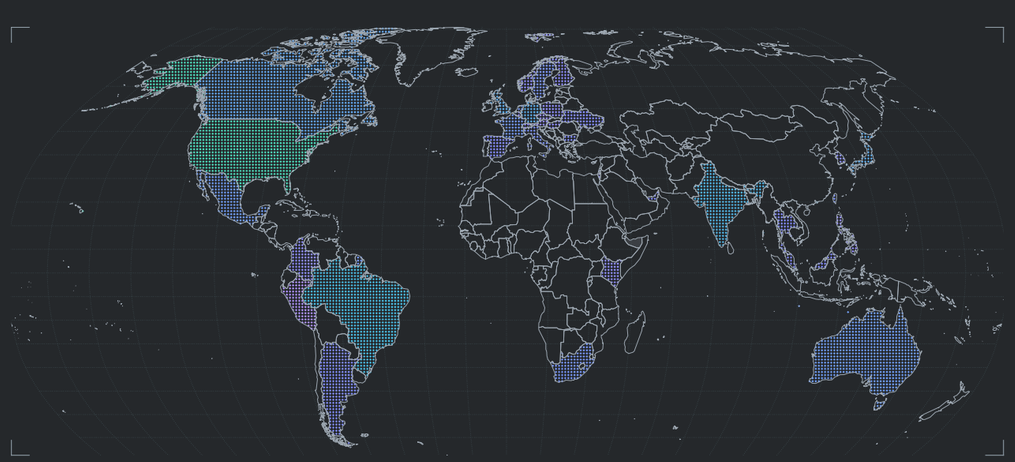
A / Amazon



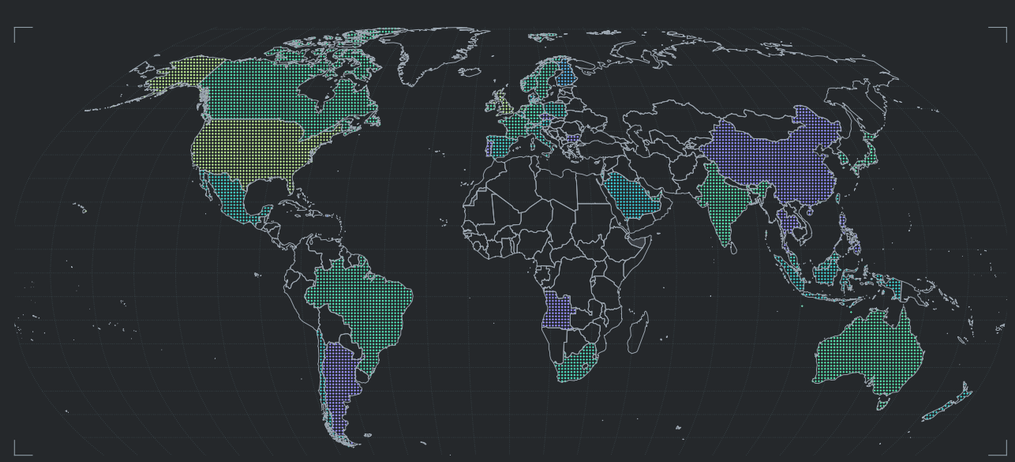
B / Meta

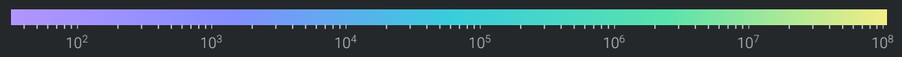


C / Google



D / Microsoft





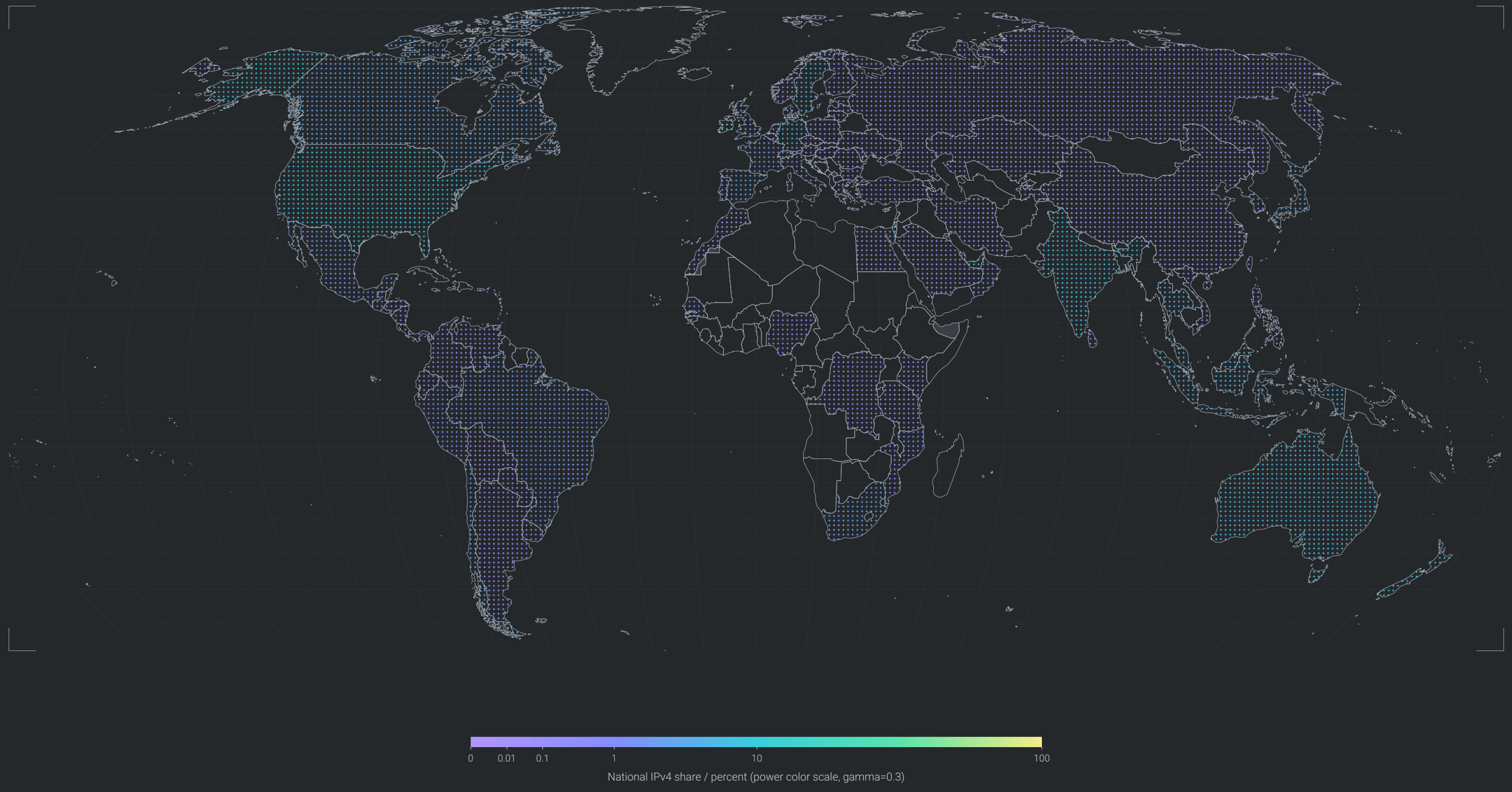

Observed IPv4 per country / logarithmic scale

METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Same BGP method for all companies.
Primary organizations only. Hatch color: national IPv4 count. Unmarked dark: zero within selection; medium grey: no denominator. Not ownership.

# Amazon / relative weight

Share of each country's observed, geolocated IPv4

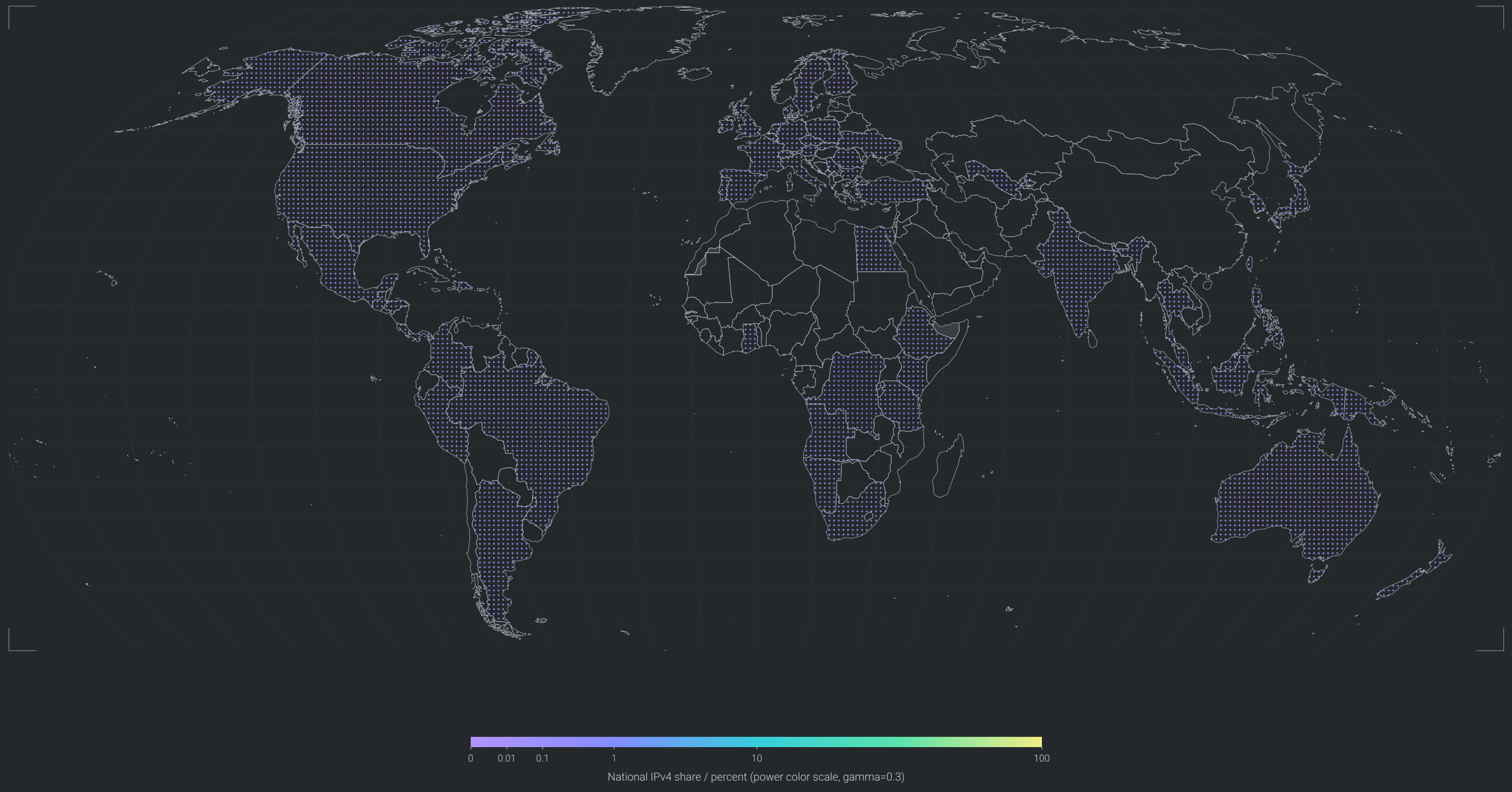



METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Hatch density is constant.
Unmarked dark: zero within selected primary organizations; medium grey: no denominator. Shares do not measure traffic, users or ownership.

# Meta / relative weight

Share of each country's observed, geolocated IPv4

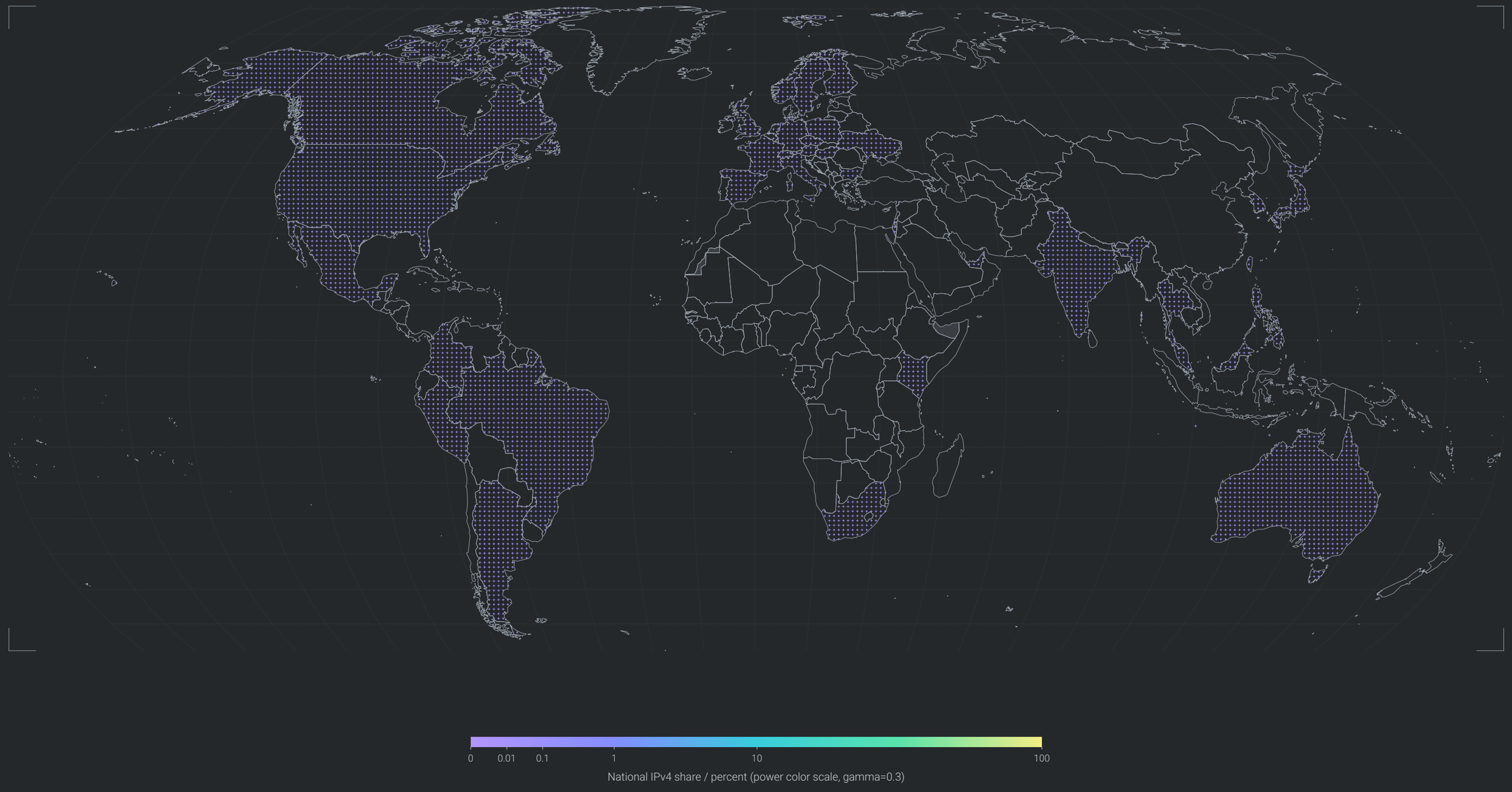



METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Hatch density is constant.
Unmarked dark: zero within selected primary organizations; medium grey: no denominator. Shares do not measure traffic, users or ownership.

# Google / relative weight

Share of each country's observed, geolocated IPv4

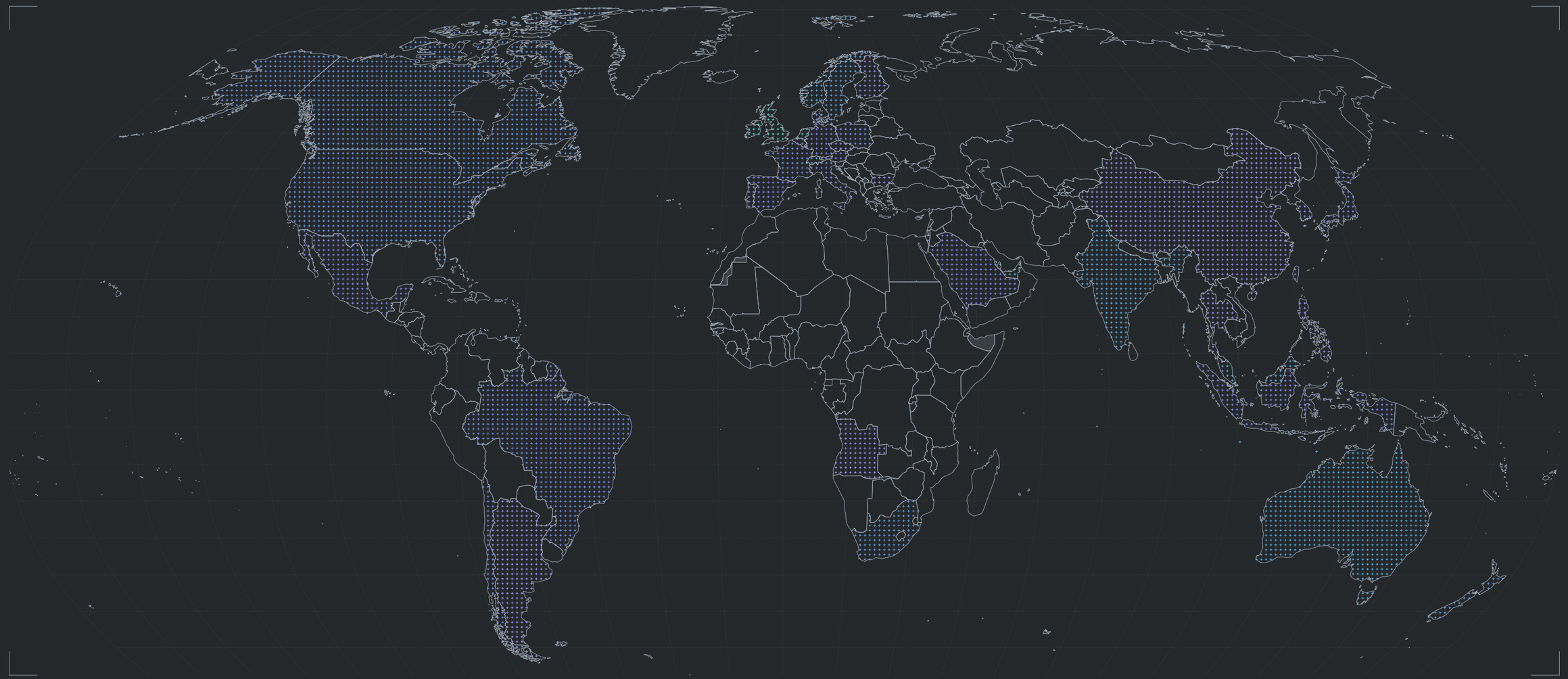



METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Hatch density is constant.
Unmarked dark: zero within selected primary organizations; medium grey: no denominator. Shares do not measure traffic, users or ownership.

# Microsoft / relative weight

Share of each country's observed, geolocated IPv4



0 0.01 0.1 1 10 100

National IPv4 share / percent (power color scale, gamma=0.3)

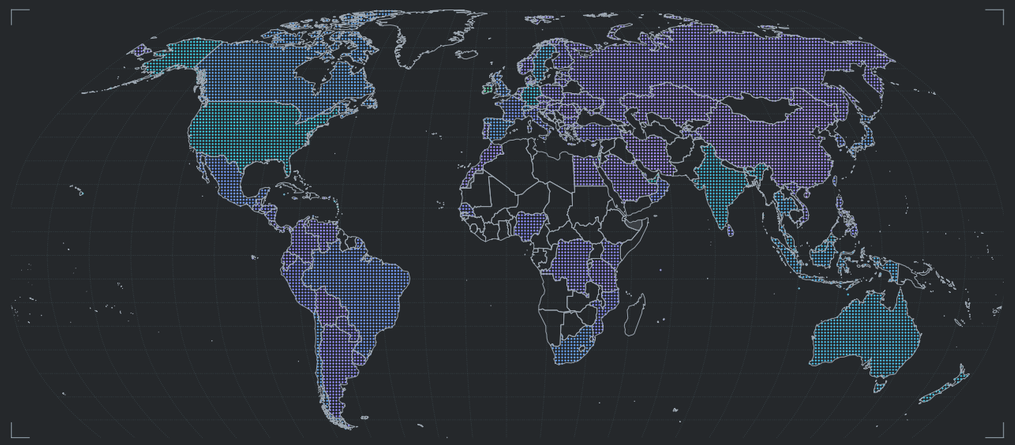
METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Hatch density is constant.
Unmarked dark: zero within selected primary organizations; medium grey: no denominator. Shares do not measure traffic, users or ownership.

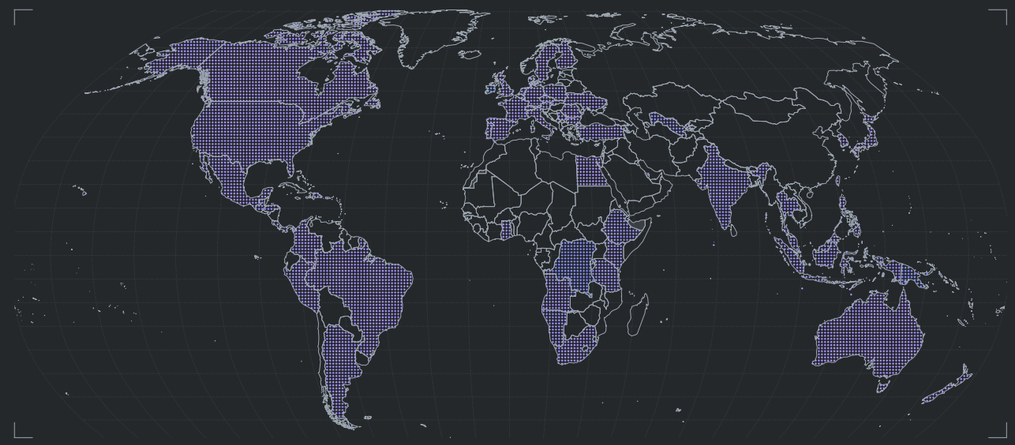
# The geography of relative weight

National shares / shared nonlinear 0-100 percent color scale

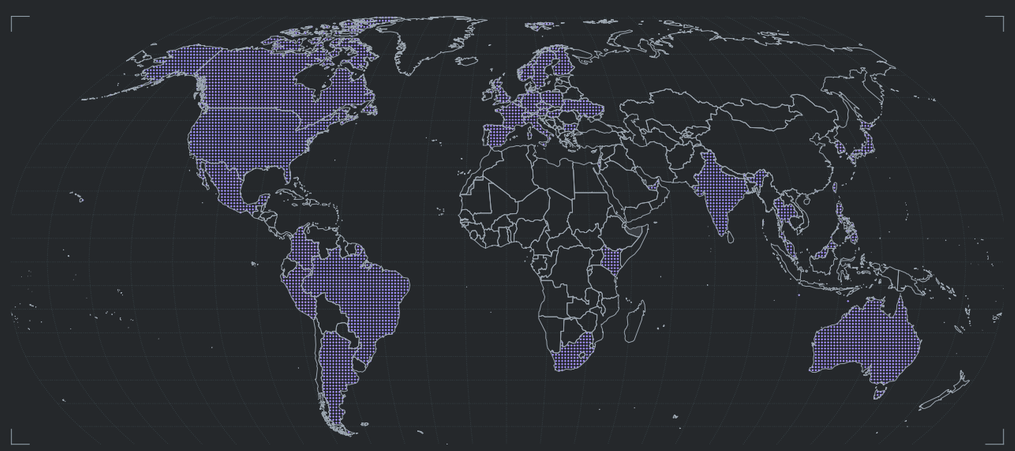
A / Amazon



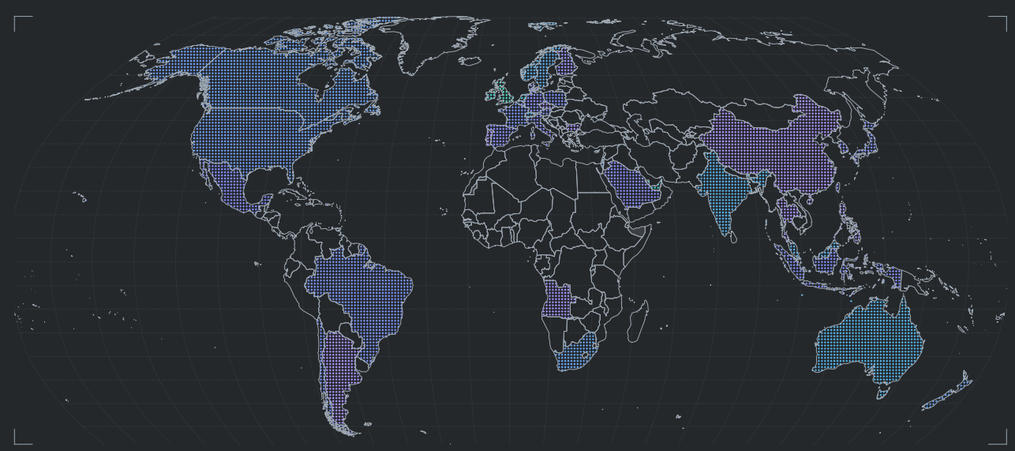
B / Meta

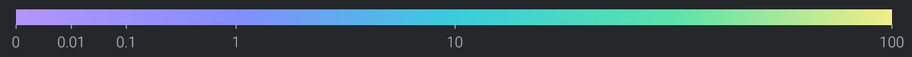


C / Google



D / Microsoft






National IPv4 share / percent (power color scale, gamma=0.3)

METHOD / SOURCE NOTES

Sources: CAIDA / RouteViews (1 October 2026); IP2Location LITE DB11; Natural Earth. Hatch density is constant.
Unmarked dark: zero within selected primary organizations; medium grey: no denominator. Shares do not measure traffic, users or ownership.